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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Subject : COMP4423 Computer Vision

Session : 2024 / 2025 Semester II

Date : 28 April 2025 Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Examiner(s): Prof. WEI Xiaoyong

This question paper has a total of 7 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

The exam is an open-book examination. Students are allowed to bring 4 pages of A4 size material to the exam.

Electronic devices are not allowed.

Students should write their answers on the answer sheet. Only the answers on the answer sheet will be graded.

Do not turn over the page until you are told to do so!

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Section C: Fill-In-Blanket Questions (20 marks)

21. In morphological operations, the process of removing small noise by eroding the boundaries of foreground objects is called _____, while the process of expanding the boundaries of foreground objects is called _____.

22. In image processing, the process of estimating pixel values at non-integer coordinates during resizing or transformation is called _____, and common techniques for this include _____, _____, and _____.

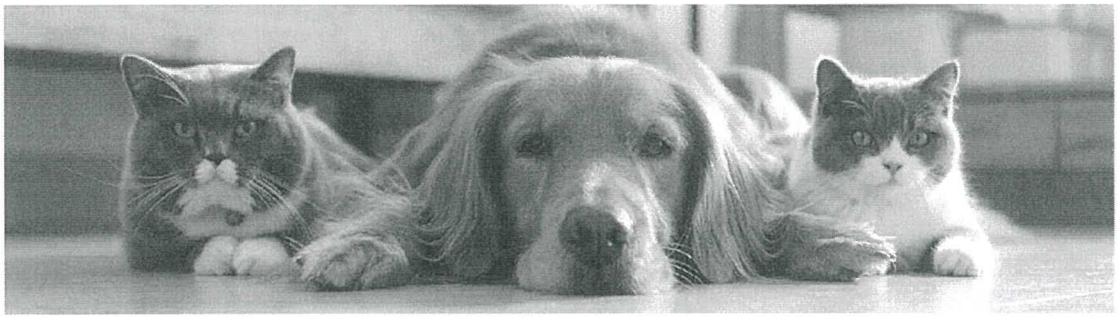
23. In Convolutional Neural Networks, the process of reducing the spatial dimensions of feature maps while retaining important information is called _____. Two common types of this operation are _____, which takes the maximum value in a region, and _____, which computes the average value in a region.

24. In neural networks, the process of computing gradients for the network's parameters by applying the chain rule of differentiation from the output layer back to the input layer is called _____. This process updates the weights using the _____ algorithm, which minimizes the _____ function.

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Section D: Short-Answer Questions (30 marks)

25. You want to monitor the activities of your two cats (Tom and Bob) and one dog (Barker) using a surveillance system powered by computer vision algorithms. The goal is to classify whether the animals are eating, sleeping, or playing and maintain a time-stamped log of their activities. Answer the following questions:



1) How would you build the time-stamped activity log? Provide a flowchart showing the necessary modules and ensure the log tracks activities for each pet (Tom, Bob, and Barker). (30 words, 5 points)

2) Which modules in your system require computer vision algorithms and models? Name the algorithms and specific models you would use for those modules. (30 words, 5 points)

3) What are the key characteristics of the dataset needed to train your models? Describe the dataset you would use in detail. (30 words, 5 points)

4) How can data augmentation enhance your system's performance? Specify at least two data augmentation techniques you would apply. (20 words, 5 points)

5) Your friend wants to use this system for their pets but has a much smaller dataset. How would you adapt your model for them? (20 words, 5 points)

6) If your model struggles with identifying animals when they are partially hidden (e.g., behind furniture), how would you improve its performance? (20 words, 5 points)

** END **